Reliance

Research Lifecycle Management technologies for Earth Science Communities and Copernicus users in EOSC

Galaxy and Research Lifecycle Management technologies for Earth Science communities and Copernicus users in EOSC

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WHO WE ARE

ANNE FOUILLOUX- https://github.com/annefou

Research Software Engineer working for the Nordic e-Infrastructure Collaboration (NeIC)

I'm currently working on:

- CodeRefinery project: it aims at promoting Best Software Practices
- NICEST2 project: I am the project leader of the second phase of the Nordic Collaboration on e-Infrastructures for Earth System Modeling.
- EOSC-Nordic project: I am involved in the WP5 on the climate Science demonstrator.
- Research Lifecycle Management technologies for Earth Science Communities and Copernicus users in EOSC (RELIANCE) INFRAEOSC-07-2020 Research and Innovation action - Grant number 101017501. 2 years project starting from January 2021.
- MAchine learning, Surface mass balance of glaciers, Snow cover, In-situ data, Volume change, Earth observation (MASSIVE) project. Funded by the Norwegian Research Council. 4 years proj.ect starting 1st September 2021 until 31 August 2025.

JEAN IAQUINTA - https://github.com/j34ni

Research Software Engineer, University of Oslo, Norway

I'm currently working on:

- RELIANCE project on the creation of Research Objects for Climate Science scenarios.
- INES (Infrastructure for Norwegian Earth System Modelling) project, project funded by the Research Council of Norway (270061) on the use of Machine Learning in Earth System Model.
- NICEST2 (the second phase of the Nordic Collaboration on e-Infrastructures for Earth System Modeling) project as the workpackage leader "Earth System Model workflows to efficiently run NorESM and EC-Earth on euroHPC and other new architectures" -

At the moment, I am developing containers (Docker, singularity) for running Earth System Model in a fully reproducible way and integrate Climate Models in Galaxy.

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RELIANCE AT A GLANCE

REsearch LIfecycle MAnagemeNt technologies for Earth ScienCe Communities and CopEmicus users in EOSC

- Reliance seeks to ***extend the EOSC's capabilities*** with an enhanced support for ***research activities*** via a set of industry-strong, innovative, ***interconnected services***, and in alignment with the ***EOSC Interoperability Framework***.
- Adopting a **holistic research lifecycle mgmt.** approach, it will ***enhance the discovery*** of and access to research data/results (incl. ***Copernicus*** data), improve the ***extraction*** of relevant information, while promoting ***FAIR and open science*** principles.
- It will pilot and demonstrate its services via ***3 Earth Science communities*** and others engaged through an ***Open Call***, fostering the use of ***Copernicus data***

Partners: PSNC, CNR ISMAR, expert.ai, MEEO, INGV, Politecnica, TerraDue, UiO Universitetet i Oslo, Alpha Consult

RELIANCE OBJECTIVES

Provision of suite of services for Research Lifecycle Mgmt.

EOSC integration and extension

Integration and onboarding of Reliance services in EOSC, leveraging and connecting with other complementary EOSC services

zenodo, EOSC-hub, OpenAIRE, eGI

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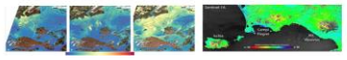
Demonstration & community engagement

Demonstrate the services and value proposition



Increase Copernicus and EOSC impact

Foster the adoption of Copernicus data and EOSC services in such scientific communities

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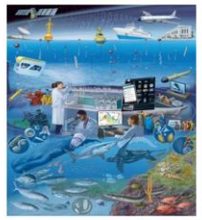
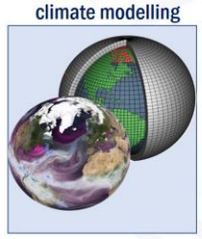
3 USER COMMUNITIES TO SHOWCASE RELIANCE

Sea Monitoring



Geohazard

Atmospheric and climate modelling



The User Communities will demonstrate how RELIANCE services can support researchers & promote the cooperation among scientists in Earth Science, though the services may be adapted to other research domain.

An open call will be open very soon to on-board new communities, including new scientific domains! Contact us if you are interested.

SERVICE PORTFOLIO

Research Lifecycle Management Ecosystem Pillars

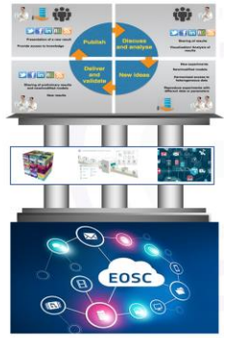
- ***Research Objects*** as the overarching mechanism to manage scientific research activities and connect associated resources

RoHub, RO-Crate

- ***Data Cubes*** for efficient and scalable structured data access and discovery

ADAM

- ***Text mining*** services to extract machine-readable metadata from RO resources enabling researchers to discover scientific information at scale and to structure their own research.



expert.ai

http://www.researchobject.org Reliance

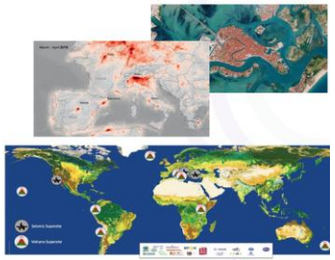
WHAT IS A DATA CUBE FOR US? AND WHY?

Collections of **geospatially-related data**, containing multi-**temporal**, multi resolution **satellite imagery** and **in-situ observations**

- First-class resources for many researchers working with Earth Observation data
- They need to be described in detail
  - Discovery
  - Archival and preservation

- Audiovisual
- Collection
- Dataset
- Event
- Image
- InteractiveResource
- Model
- PhysicalObject
- Service
- Software
- Sound
- Text
- Workflow
- Other

From the OpenAIRE dataset types....
- Which ones should we adopt?
- How much metadata can be extracted from them?



OpenAIRE dataset types

OpenAIRE

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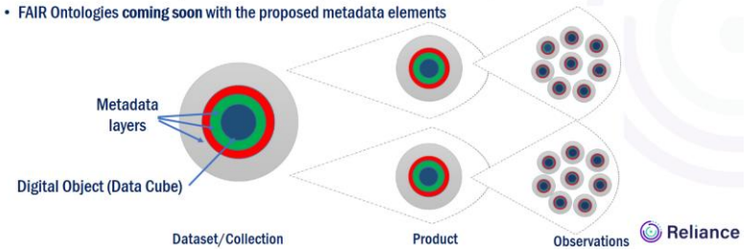
A PROPOSAL FOR A METADATA MODEL

- Considering crosswalks across the previous metadata models, and focusing on a layered approach (4 layers)
  - **General** metadata layer
  - **Credit**, licensing and pricing layer
  - **Coverage** metadata layer: spatial, temporal, vertical
  - **Data acquisition** and product information layer: provenance, variables, etc.
- FAIR Ontologies **coming soon** with the proposed metadata elements

Available at RELIANCE deliverable D4.1
https://box.pionier.net.pl/d/529599305c0e43d9bf28/



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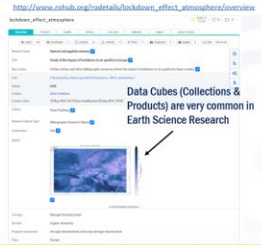
EARTH SCIENCE RO-CRATE PROFILE

We create the concept of **Data Cube Data Entity**

Relevant metadata for a Data Cube

- Simple **geolocation** (sch:contentLocation) and **temporal coverage** (sch:temporalCoverage)
- **Spatial coverage** (extent, resolution, crs)
- **Temporal coverage** (startTime, endTime, resolution-step)
- **Vertical coverage** (highestElevation, lowestElevation, resolution-step)
- **Other** (processing level, processor, instrument, platform, etc.)

http://www.rohub.org/rodetails/lockdown_effect_atmosphere/overview



A valid RELIANCE *RO-Crate JSON-LD* graph MUST describe:
1. The RO-Crate Metadata File Descriptor
2. The Root Data Entity
3. One or more Data Cube Data Entities
4. Zero or more Data Entities
5. Zero or more Contextual Entities

Reliance, RO-Crate

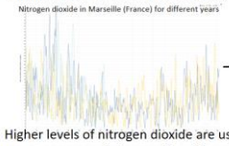
RELIANCE & GALAXY FOR THE ATMOSPHERE & CLIMATE MODELLING COMMUNITY

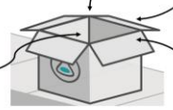
I use my ORCID to login to RoHub

Then I create a new Research Object in RoHUB at the very beginning of my project. I start adding links to my bibliography and any other material useful for my research.

For instance, I would like to study the impact of the lockdown and reduced human activity on the air quality

Create visualization in Galaxy, share and add links to my live Research Object.

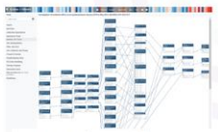




Higher levels of nitrogen dioxide are usually observed in winter. So never compare winter months with spring/summer months as it has been shown many times for explaning the reduction in NO2 over the lockdown!

I use Galaxy for analyzing Copernicus data and link my Galaxy history to my live Research Object

https://youtu.be/FjDL-yWdhD4

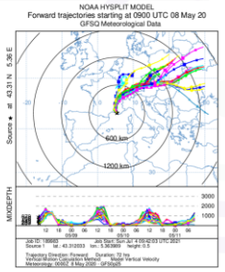


I create a Galaxy workflow to create timeseries so I can re-run the same analysis for different locations..

WHAT OTHER TOOLS DO WE NEED?

To properly study the impact of lockdown or the reduction in human activities on the air quality, we need to take into account the meteorological conditions such as the wind (intensity and direction), temperature, pressure, sunlight, etc.

Run HYSPLIT (trajectory model developed and maintained by NOAA, US) to find out where the pollutants are coming from or where they are going to.



Run Atmospheric dispersion models such as EMEP MSC-W model with «if» scenarios for instance to improve models, develop plausible mitigation strategy.

The EMEP MSC-W model is designed to calculate air concentrations and deposition fields for major acidifying and eutrophying pollutants, photo-oxidants and particulate matter. The Open Source releases are intended to:

- facilitate insight in the model assumptions, the parametrisation used, the requirements for input data and the actual model code
- encourage dialogue and collaboration with the modelling community
- allow individual model runs and insight on how to run different scenarios

See https://emep.int/mscw/

GALAXY & RELIANCE – NEXT STEPS

Climate Science Workbench

- Add Data Cube in Galaxy using the metadata model proposed by RELIANCE
- Import/export Galaxy histories to RoHUB and vice versa
- Add new tools for our use cases such as trajectory models (HYSPLIT; FLEXPART, LAGRANTO), atmospheric dispersion models (EMEP MSC-W) and develop workflows
- Understand how to use Galaxy Community registry for creating collaboration sphere in Rohub and connect Galaxy users based on their histories (Research Objects) and centres of interest.

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